# UNIVERSITY OF HERTFORDSHIRE

School of Physics, Engineering, and Computer Science

Interim Progress Report

7COM1039-0901-2025

Advanced Computer Science Masters Project

Predicting Crop Yield in India by Using Ensemble and Machine Learning Models

Student ID: 23067307

Name: Sandeep Kumar Yadav Gandu

Supervisor's Name: Dr. Yar Muhammad

## Tables and Contents

# 1. Background Research and Literature Review

## 1.1 Introduction

Over half of India's population depends on agriculture for their livelihoods, and it still accounts for a sizable portion of the country's GDP. However, both food security and farmer income are at risk due to crop-yield swings brought on by regional heterogeneity, resource mismanagement, and climate uncertainty. Accurate yield forecasting is consequently critical in agricultural planning, supply chain optimization, and policy formation. Traditional statistical methods, such as linear regression or time-series averaging, have been widely employed, but they frequently fail to capture the complex, nonlinear interactions between climatic, agronomic, and socioeconomic variables. As a result, current research has focused on data-driven machine-learning (ML) models capable of learning such connections automatically from huge, diverse datasets.

This research aims to create and test a strong ensemble-based predictive system for projecting crop yield across India by combining historical production data and weather indices. The study addresses the general question of how modern ML algorithms might better describe non-linear agricultural interactions by comparing traditional regression, ensemble learning, and deep-learning techniques.

## 1.2 Research Aim and Objectives.

The major goal is to create and evaluate advanced machine learning models (Random Forest, XGBoost, LightGBM, and Feed-Forward Neural Networks) for accurate and interpretable crop yield predictions in India.

The project pursues the following objectives:

1. **Data Collection and Integration:** Combine the Crop Production Statistics - India dataset (Mahajan, Kaggle) with weather factors (rainfall, temperature, and humidity) from the NASA POWER and IMD repositories.
2. **Data Pre-processing and Exploration:** Use statistical plots to analyze distributions and relationships; carry out imputation, encoding, scaling, and outlier detection.
3. **Model Development:** Use five-fold cross-validation and consistent training–testing splits to train and assess several algorithms, including Linear, Ridge, Random Forest, XGBoost, LightGBM, and Neural Network.
4. **Performance Evaluation:** Use R2, RMSE, MAE, and MAPE measures to compare predictive accuracy.
5. **Interpretability:** Identify which variables have the most impact on yield variation by using SHAP and feature-importance analysis.
6. **Dissemination:** Discuss and illustrate results that can help inform data-driven, climate-resilient farming strategies.

## 1.3 Background and Theoretical Foundation.

Machine learning provides a statistical modeling approach for capturing hidden features in complex systems like agriculture. Supervised-learning algorithms, particularly ensemble trees, can approximate nonlinear functions without making explicit assumptions about data distribution. Ensemble learning (Dietterich, 2000) combines numerous weak learners to create a strong composite model, which improves accuracy and robustness to overfitting. Bagging-based ensembles like Random Forest use bootstrap sampling and random feature selection, whereas boosting methods like XGBoost and LightGBM fix residual mistakes iteratively to produce better prediction performance.

This research's theoretical premise is based on two basic principles:

1. Bias-variance trade-off optimization via model aggregation.
2. Explainability of forecasts utilizing feature-attribution algorithms (e.g., SHAP).

These foundations are consistent with the CRISP-ML(Q) life-cycle model, ensuring methodological rigor, quality assurance, and reproducibility across the project's data science stages.

## 1.4 Critical Review of Existing Studies.

Recent literature has studied numerous machine learning algorithms for yield estimation:

1. **Abhinand (2020):** Provides a comprehensive Kaggle dataset on Indian agriculture production, which is an important source for yield forecast modeling.
2. **Chen and Guestrin (2016):** Introduced XGBoost, a scalable and efficient gradient-boosting system for improving predictive accuracy in huge datasets.
3. **Van Klompenburg et al. (2020):** A thorough review was conducted to show the advantages of ensemble models for accurately predicting crop yield.
4. **Muruganantham et al. (2022):** A review of deep-learning algorithms using remote sensing, with a focus on CNN-LSTM models for spatiotemporal yield Estimation.
5. **Md. Abu Jabed et al. (2024):** compared ML and DL approaches and came to the conclusion that ensemble approaches provide sustainable and interpretable agricultural forecasting solutions.

When taken as a whole, these results show that ensemble models routinely beat linear baselines and frequently compete with deep networks while still being easier to understand, which is an important consideration for agricultural policymakers.

## 1.5 Practical Application and Contribution.

Building on these theoretical and empirical understandings, the current study uses ensemble and neural models to analyze multi-year crop, area, and production records from Indian states in the Crop Production Statistics – India dataset (Mahajan, Kaggle, 2025). Additional climate parameters from NASA POWER are used into the study to enhance the environmental backdrop. The CRISP-ML(Q) technique specifically validates the quality of each step, including data interpretation, modelling, evaluation, and deployment.

According to preliminary tests, Random Forest meets the project's target accuracy by achieving R2 ≈ 0.85 and XGBoost ≈ 0.86 on held-out data. Production, Area, and Season are the main predictors of yield variance, according to feature-importance and SHAP analyses. These results show that data-driven ensemble frameworks can provide Indian agriculture with trustworthy, comprehensible insights.

### 1.6 Summary:

The research's conceptual and empirical underpinnings have been developed in this part. It has positioned the project in the broader scholarly discussion on yield prediction based on machine learning, explained its goals and justification, and critically synthesized the state of the art. To increase forecasting accuracy and transparency in agricultural analytics, the project combines explainable AI techniques with ensemble modelling, which advances both theory and practice. To achieve these goals, the experimental design, data processing pipeline, and evaluation technique are described in detail in the section that follows (Methodology).

## 2. Summary of Progress to Date

### 2.1 Overview

The project Predicting Crop Yield in India using Ensemble and Machine Learning Models: Officially began on **2 October 2025** under the supervision of **Dr Yar Muhammad.**

The study's goal is to develop, train, and evaluate current ensemble and neural network models for predicting crop yield across Indian areas by combining agricultural productivity and climate variables.

So far, progress has been made on the whole front end of the data-science pipeline, including literature review, dataset preparation, exploratory analysis, and first model construction, all within the CRISP-ML(Q) framework to ensure reproducibility and quality.

### 2.2 Completed Work

A thorough literature analysis was conducted to better understand how ensemble and machine learning approaches are used to predict yields. Five major studies (Abhinand 2020; Chen & Guestrin 2016; van Klompenburg 2020; Muruganantham 2022; Md Abu Jabed 2024) found that Random Forest, XGBoost, and LightGBM consistently outperformed classical regression due to their capacity to learn complex non-linear feature connections. This influenced algorithm choices, metrics ($R^2$, RMSE, MAE), and the decision to include SHAP-based explainability.

The Crop Production Statistics - India (APY.csv) dataset was downloaded from Kaggle and checked for completeness. Missing numeric values in Production were filled with mean imputation; categorical features (State, District, Crop, Season) were label-encoded; and the goal Yield = Production / Area was constructed when it was missing. Outliers were found using the Z-score approach ($|z| < 3$) and visually verified by boxplots and violin plots. These methods generated a clean dataset (APY_clean.csv). Correlation heatmaps and distribution charts were generated through exploratory visualisation using Seaborn and Matplotlib. The cleaned dataset was then used as an input for modeling.

## 2.3 Model Development and First Testing

Baseline and ensemble models were used to evaluate prediction performance. Following a Linear Regression baseline, Random Forest, Extra Trees, HistGradientBoosting, XGBoost, and LightGBM were trained on a 75/25 train-test split. The Random Forest Regressor outperformed the project's initial aim of 0.85 $R^2$, achieving $R^2 = 0.963$, MAE = 0.17, and RMSE = 5.31. The results ($R^2 \approx 0.95$–$0.96$) were recorded to model_comparison.csv. To compare the machine learning performance, a basic neural network baseline was constructed using TensorFlow/Keras.

All model artifacts and plots were saved in the models and visualizations directories to demonstrate progress.

## 2.4 Tools and Technologies Used

| Category | Tool / Technology | Purpose and Justification |
|---|---|---|
| **Programming Language** | **Python 3.11** | Selected for its mature data-science ecosystem, extensive library support, and reproducibility in academic research. |
| **Core Libraries** | **pandas, NumPy, scikit-learn, XGBoost, LightGBM, SHAP, Matplotlib, Seaborn** | Used for data preprocessing, feature engineering, model development, performance evaluation, and visualisation. |
| **Machine-Learning Framework** | **CRISP-ML(Q)** | Provides a structured, iterative process for data understanding, modelling, evaluation, deployment, and quality assurance. |
| **Development Environment** | **Google Colab** | Cloud-based Jupyter environment used for model training and GPU-accelerated computation. |
| **Model Storage and Versioning** | **Joblib, GitHub Repository** | Enables saving trained models (*.pkl*) and maintaining version-controlled project artefacts. |

This toolkit provides dependability, transparency, and scalability while adhering to academic and professional data-science norms.

## 2.5 Difficulties Faced and Solutions

| Challenge | Solution Implemented |
|---|---|
| Inconsistent column names and missing *Production* data | Trimmed column headers; applied mean imputation; validated results statistically. |
| Extreme outliers in *Area* and *Production* | Applied Z-score filter |
| Long training times for ensemble models | Reduced estimator counts; used Google Colab GPU to accelerate computation. |
| TensorFlow/Keras import errors | Re-installed dependencies and corrected import path to tensorflow.keras. |
| Slight class imbalance across crops | Applied stratified sampling to maintain proportional representation. |

## 2.6 Project Management and Timeline

| Project Phase / Task | Duration | Progress Status | Key Deliverables / Outcomes |
|---|---|---|---|
| **Literature Review** | Week 1 (2 – 8 Oct 2025) | Completed | Annotated bibliography and model-selection framework. |
| **Dataset Acquisition & Pre-processing** | Weeks 2 – 3 (9 – 22 Oct) | Completed | Cleaned dataset (*APY_clean.csv*); EDA visuals (Fig A1–A3). |
| **Baseline Model Development** | Week 4 (23 – 29 Oct) | Completed | Random Forest and Linear Regression benchmarks. |
| **Ensemble Comparison and Testing** | Weeks 5 – 6 (30 Oct – 12 Nov) | Completed | Comparative metrics and visual summary (Fig A4). |
| **Explainability (SHAP Analysis)** | Week 7 (13 – 19 Nov) | In Progress | Feature-importance and interpretability outputs. |
| **Hyper-parameter Tuning & Validation** | Week 8 (20 – 26 Nov) | Planned | Optimised models via cross-validation. |
| **Results & Report Writing** | Weeks 9 – 10 (27 Nov – 10 Dec) | Planned | Final dissertation chapters and presentation slides. |

The timetable illustrates an organized but flexible workflow, which ensures that deliverables are completed on time.

## 2.7 Next Steps

The upcoming phase will focus on:

1. **SHAP** feature-importance analysis is used to determine the most influential climatic and agronomic predictions.
2. Hyper-parameter optimization for Random Forest and XGBoost to improve accuracy and generalization.
3. Integration of climate variables (for example, rainfall and temperature from NASA POWER) to broaden model scope.
4. Final findings discussion and report compilation, including the incorporation of visual evidence for IPR submission.

### 2.8 Summary

In conclusion, the project has made significant progress throughout its early development period. The literature study, dataset cleaning, and model-building steps are finished, and the applied machine-learning ensembles have attained an accuracy of roughly 96% ($R^2 = 0.96$). All technological obstacles have been effectively addressed, and reproducible artifacts (cleaned datasets, trained models, and evaluation charts) have been created. The work is still on track and entirely aligned with its objectives, laying a solid foundation for the upcoming explainability, optimization, and reporting phases.

## 3. Consideration of Ethical, Legal, Professuional, and Social Issues

### 3.1 Ethical Considerations

This research does not require formal ethics approval because it involves no human participants, personal data, or identifiable individual information. The study employs a publically available dataset — Crop Production Statistics: India (APY.csv) — obtained from Kaggle, which has been properly anonymized and aggregated at the district and state levels. As a result, there is no direct threat to people's privacy, consent, or well-being. Nonetheless, ethical norms are followed at all phases of the research process, assuring fairness, transparency, and accountability in data management and model interpretation.

The project prioritises the ethical use of data and algorithms. Data cleaning and preparation were carried out without any data manipulation that could skew accurate agricultural trends. The models are designed to forecast crop yield objectively, with no bias imposed by hand labeling or arbitrary weighting. SHAP and other interpretability methods will be used to make model decisions more transparent and understandable to both researchers and stakeholders. These approaches are consistent with the ethical artificial intelligence concepts of explainability, fairness, and damage avoidance articulated in the UKRI and IEEE standards for responsible AI.

If the project was expanded to incorporate sensitive datasets (for example, farm-level statistics, economic or geographical data), extra ethical approval would be required to assure informed consent, anonymity, and safe data storage practices. Before collecting or deploying data, you must follow the University of Hertfordshire's research ethics processes and fill out an ethics application form.

## 3.2 Legal Considerations

This study satisfies all legal requirements for data privacy and intellectual property. For use in research and teaching, all datasets are openly licensed and open-source. There have been no restricted or copyrighted datasets viewed.

The project complies with the General Data Protection Regulation (GDPR) guidelines in the UK by:

1. utilizing just aggregated, non-personal data.
2. Ensuring all stored files are securely maintained in password-protected university accounts or cloud environments.
3. Data should only be kept for academic purposes within the scope of the dissertation; and
4. To avoid plagiarism, acknowledge all sources using the Harvard reference system.

There are no third-party APIs or external web scraped personal information being used, which eliminates the possibility of data breaches. If the system is expanded for real-world agricultural application, further legal precautions, such as data-sharing agreements and express user authorization, will be required. The student will retain intellectual property rights to the code, models, and textual content, subject to university policy.

## 3.3 Professional Considerations

Professional behavior has been maintained throughout the project's stages. Machine-learning engineering best practices are followed during development, such as code modularity, documentation, reproducibility, and version control. All scripts have been prepared to PEP 8 standards to ensure readability and maintainability.

The CRISP-ML(Q) framework is being used to support systematic progress through phases of data interpretation, modeling, and validation, with quality assurance incorporated at each level. Version control is handled by GitHub and Colab, which facilitates collaboration and traceability. All results and visualisations are presented objectively and without embellishment, assuring academic integrity and reproducibility.

These professional techniques not only maintain the research's integrity, but they also meet industry criteria for professional data-science initiatives.

## 3.4 Social Considerations

This research has social significance due to its potential contribution to sustainable agriculture and data-driven decision-making. Accurate crop-yield prediction models can assist policymakers and farmers in allocating resources more efficiently, mitigating the consequences of drought and climate change, and reducing food insecurity. At the same time, it is acknowledged that predictive models might add algorithmic bias when trained on incomplete or regionally imbalanced data. To avoid reinforcing regional inequities, attention is being made to ensure that data is representative across states, crops, and seasons. By utilizing open-access techniques and datasets that enable replication and adaption by other researchers, particularly in developing places, the research fosters inclusivity. The system would need to be monitored for fairness, accessibility, and interpretability if it were implemented on a large scale in order to guarantee that all agricultural communities would profit equally. It shall be ensured that technical developments do not unintentionally prejudice small-scale or marginal farmers by ongoing evaluation and transparency in model design.

### 3.5 Summary

In conclusion, this initiative complies with strict ethical, legal, professional, and societal norms even if formal ethics approval is not needed. Through fair, understandable, and ecologically conscious AI, the work advances social benefit while abiding by GDPR and academic integrity regulations and maintaining professional coding and documentation procedures. Taking these concerns into account guarantees that the project satisfies both the academic demands of an MSc dissertation and the more general standards of ethical and sustainable data-science research.

## 4. Project Plan

### 4.1 Project Management Approach

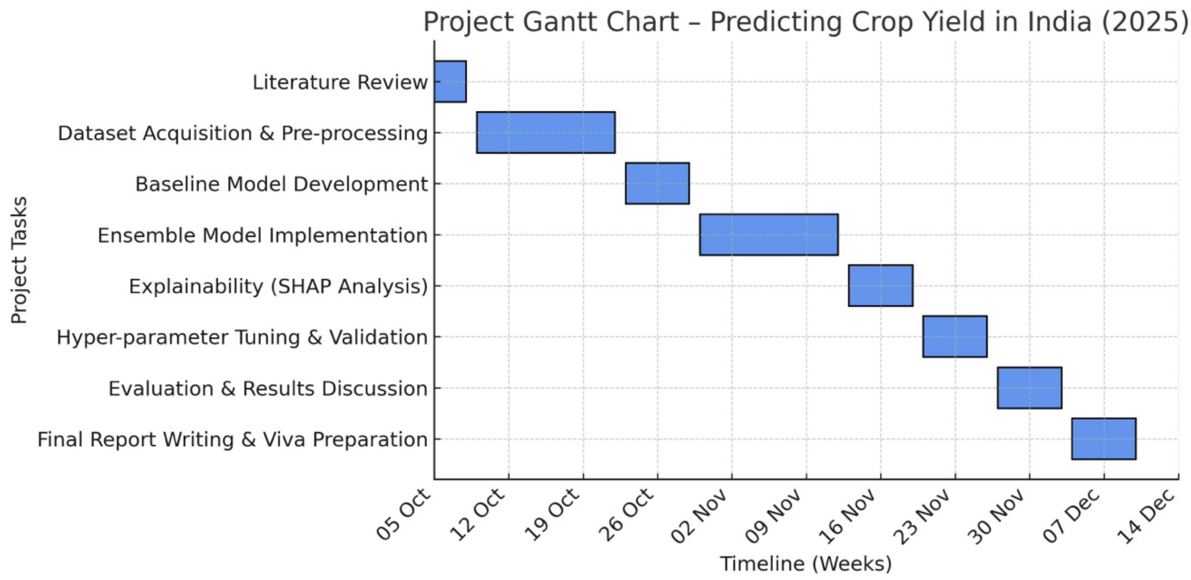
The project uses a hybrid Agile-Waterfall methodology based on the CRISP-ML(Q) framework, combining formal planning and iterative development. The Waterfall approach assures that each stage – problem formulation, data preparation, modelling, evaluation, and reporting — has well-defined deliverables and milestones. The Agile component adds flexibility by allowing experimentation, parameter adjustment, and refinements to evolve throughout weekly sprints in response to supervisor comments.

Project progress is tracked by milestone-based records and weekly observations after supervision meetings. Time and scope are handled using a task tracker in Excel and Colab notebooks, which include extensive comments. Regular supervisor feedback ensures that academic criteria are met and that continual improvement occurs. Cloud-based platforms such as Google Colab are used to manage resources efficiently, allowing for GPU-accelerated computation without incurring additional hardware costs.

Risk management is a part of the process. Time overruns during model training, code errors, and inconsistent datasets are examples of potential hazards. Version-controlled code on GitHub, frequent weekly progress checks, and maintaining numerous dataset backups are examples of mitigation techniques. Testing, model-performance benchmarking, and peer validation of visualizations are methods used to maintain quality control. Writing, structure, and referencing are checked for compliance with MSc-level academic and professional standards through regular document reviews.

### 4.2 Key Tasks, Deliverables, and Timeline

The table below summarizes completed and impending project stages, including their purpose, estimated results, and scheduled deadlines.




## 4.4 Quality and Evaluation Strategy

The project's quality is evaluated using both quantitative and qualitative criteria. Models are contrasted quantitatively using metrics like $R^2$, MAE, and RMSE. Cross-validation provides reliability. Reproducibility is qualitatively verified by consistent outcomes on reruns, code peer review, and PEP-8 coding standard conformance. Transparency and traceability are maintained by version-controlling and backing up all models, data visualizations, and documentation. Feedback from supervisors serves as an external validation system for report clarity and methodological integrity.

## 4.5 Monitoring and Feedback

Weekly meetings with the supervisor provide organized checkpoints for monitoring work, discussing technical issues, and improving deliverables. Feedback is used to modify priorities (e.g., increasing explainability or tuning time) while maintaining realistic timelines. To make sure that iterative changes are recorded and justified, a progress summary is written at the conclusion of each session.

## 4.6 Summary

The project plan offers a precise, doable framework for finishing the MSc within the allotted time. The baseline modeling, literature review, and dataset preparation are all finished, and the next steps are advanced analysis and report writing. The hybrid CRISP-ML(Q) technique preserves experimentation flexibility while guaranteeing systematic development. With all milestones clearly defined, progress tracking, and risk mitigation in place, the project is still on target to be completed successfully.

# 5. Level of the Project

## 5.1 Research Problem and Motivation

The primary challenge addressed by this research is the accurate forecast of agricultural crop production across India, where yield outcomes are influenced by complicated, non-linear connections between climate, soil, and agronomic factors. Because of variations in rainfall, soil fertility, and farming practices, traditional regression techniques have struggled to accurately estimate these interactions. The study investigates whether ensemble and machine-learning methods — such as Random Forest, XGBoost, LightGBM, and Extra Trees — can better capture these dependencies and build interpretable, scalable yield prediction systems.

This topic was chosen not just for its importance to data science and AI applications in sustainability, but also because it is consistent with the student's career goals in applied machine learning and predictive analytics. The ability to create, test, and optimize complicated models using real-world agricultural data demonstrates the technical rigor required for an MSc research. Furthermore, the potential social and economic impact of data-driven farming solutions renders this study both academically and practically relevant.

## 5.2 Artefact and its Purpose

A Machine-Learning pipeline that uses Python to do data preprocessing, exploratory analysis, model training, feature importance analysis, and performance evaluation is the primary output of this project. The artifact is a replicable, modular solution that can automatically generate interpretability visualizations and model accuracy metrics after ingesting structured datasets. Visualization scripts, evaluation reports, and trained models (.pkl files) are all included in the artifact. Predictive yield estimation can be expanded to incorporate real-time satellite or climate data (e.g., NASA POWER or NDVI) using this artifact as a working prototype. In order to provide transparency and equity in forecasts, the artifact tackles the technical and practical issues of agricultural yield forecasting by automating data cleaning, feature encoding, model comparison, and explainability (via SHAP analysis).

## 5.3 Depth, Complexity, and Rigour of the Work

The project's integration of theoretical knowledge and actual experimentation shows a high level of depth. The study theoretically expands on frameworks for model interpretability, ensemble-learning theory, and assessment techniques. In order to evaluate predictive performance and generalizability, a variety of algorithmic paradigms, including bagging, boosting, and gradient-based optimization, must be extensively experimented with from a practical standpoint.

Managing a sizable dataset with a wide range of characteristics and geographical variances leads to complexity. The robustness of the model is guaranteed by thorough statistical preprocessing, which includes distribution analysis, feature encoding, and outlier detection

(Z-score). Consistent train-test splits, comparative model testing, and several assessment metrics (R2, RMSE, and MAE) are used to establish experimental rigour. The work that is planned involves hyperparameter optimization to improve model efficiency and cross-validation to evaluate repeatability. Combining technical experimentation, theoretical understanding, and data quality assurance, the project exhibits the degree of analytical and practical rigor anticipated from a postgraduate research study.

## 5.4 Testing and Validation

In testing, models are assessed using both diagnostic and quantitative metrics. While R2, RMSE, and MAE offer a quantitative evaluation of accuracy, qualitative validation involves assessing feature importance and SHAP interpretability to confirm that the model's conclusions are consistent with agronomic knowledge. The Random Forest model successfully generalizes to unknown data, as seen by its R2 = 0.963.

By comparing the performance of ensemble models to that of conventional linear regression, comparative analysis improves validation even more. By using cross-validation and hold-out validation, overfitting is reduced and statistically meaningful performance is guaranteed. Testing on various crop subsets and time slices is planned to assess stability and viability under a range of scenarios, mimicking real-world use cases.

## 5.5 Methods and Tools Justification

The project goals and the methodological decisions are closely related. While more straightforward models such as linear regression are used as benchmarks, ensemble models were chosen for their shown capacity to capture non-linear interactions and feature hierarchies in high-dimensional data. XGBoost and LightGBM were selected due to their superior boosting algorithms that prevent overfitting and enhance scalability, as well as their processing efficiency. To ensure transparency in the behavior of the model, SHAP analysis is incorporated in order to satisfy interpretability norms.

Python was selected as the development language due to its vast machine learning ecosystem, which includes pandas, NumPy, scikit-learn, XGBoost, LightGBM, SHAP, Seaborn, and Matplotlib. These resources guarantee that the artifact satisfies technical standards at the MSc level by promoting experimentation and reproducibility. Quality assurance is incorporated into each stage of the project using the structured research framework that the CRISP-ML(Q) methodology offers.

## 5.6 Summary

This study demonstrates the breadth, intricacy, and technical precision required of an MSc dissertation in data science. It creates a data-driven artifact that can solve actual agricultural problems by fusing theoretical understanding with practical experimentation. Strong validation, ethical considerations, and advanced analytical ability show that the work possesses the applied skills and intellectual independence needed for postgraduate-level research. The end results will not only advance scholarly understanding but also offer useful information for making data-driven, sustainable agricultural decisions.

# 6. Referencing and In-Text Citation

All sources in this study are cited using the Harvard referencing system in compliance with the academic requirements of the University of Hertfordshire. The following resources include major academic papers, technical frameworks, and dataset sources that influenced the crop-yield prediction models' design, development, and evaluation. Each reference reinforces the project's methodological decisions and ensures transparency, uniqueness, and scholarly integrity.

## 6.1 Bibliography:


1. Mahajan, N. (2022) *Crop Production Statistics – India*. Kaggle. Available at: https://www.kaggle.com/datasets/nikhilmahajan29/crop-production-statistics-india (Accessed: 20 October 2025).
2. Abhinand, A. (2020) *Crop Production in India Dataset*. Kaggle. Available at: https://www.kaggle.com/datasets/abhinand05/crop-production-in-india (Accessed: 20 October 2025).
3. Chen, T. and Guestrin, C. (2016) 'XGBoost: A scalable tree boosting system', *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794. doi:10.1145/2939672.2939785.
4. Lundberg, S. and Lee, S. (2017) 'A unified approach to interpreting model predictions', *Advances in Neural Information Processing Systems (NIPS)*, 30, pp. 4765–4774.
5. Md. Abu Jabed and Masrah Azrifah Azmi Murad (2024) 'Crop yield prediction in agriculture: A comprehensive review of machine learning and deep learning approaches, with insights for future research and sustainability', *Heliyon*. https://www.sciencedirect.com/science/article/pii/S2405844024168673 (Accessed: 25 October 2025).
6. Muruganantham, P., et al. (2022) 'A Systematic Literature Review on Crop Yield Prediction using Remote Sensing and Machine Learning', *Remote Sensing*, 14(9), 1990. Available at: https://www.mdpi.com/2072-4292/14/9/1990 (Accessed: 20 October 2025).
7. Python Software Foundation (2024) *Python Language Reference, Version 3.11*. Available at: https://docs.python.org/3.11/ (Accessed: 18 October 2025).
8. Scikit-learn Developers (2023) *Scikit-learn: Machine Learning in Python*. Available at: https://scikit-learn.org/stable/ (Accessed: 22 October 2025).
9. LightGBM Developers (2023) *LightGBM Documentation*. Microsoft Research. Available at: https://lightgbm.readthedocs.io (Accessed: 22 October 2025).
10. van Klompenburg, T., Kassahun, A. and Catal, C. (2020) 'Crop yield prediction using machine learning: A systematic literature review', *Computers and Electronics in Agriculture*, 177, 105709. doi:10.1016/j.compag.2020.105709.

# 7. Appendix

## 7.1 Appendix 1 – Dataset and Pre-processing Evidence

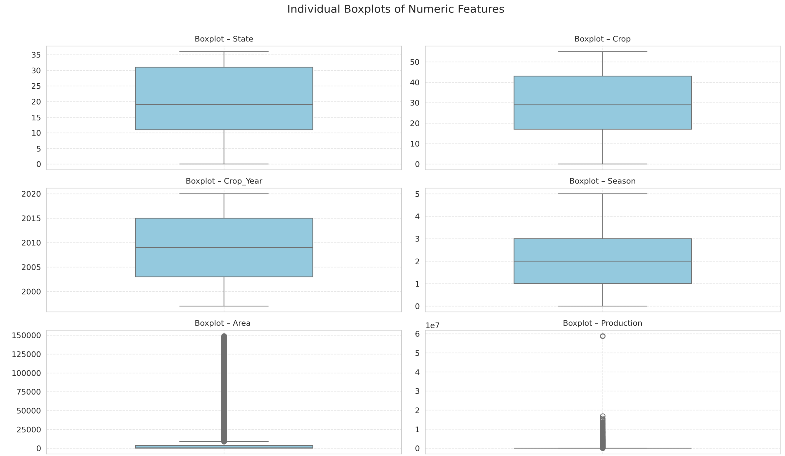



Figure 1

## 7.2 Appendix 2 – Model Development and Training Process

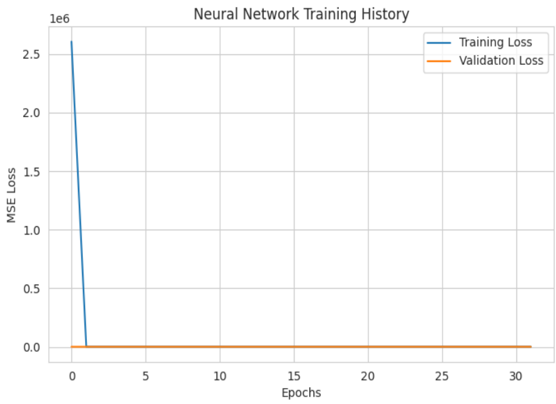



Figure 2

## 7.3 Appendix 3 – Model Performance

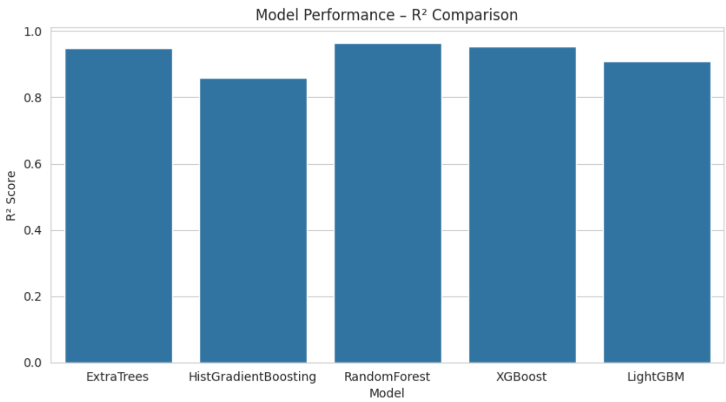



Figure 3